AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I067*

Team Name: *Aviators*

Domain: *Space, Aerospace, Mobility and Next-Gen Infrastructure*

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Problem Statement

Commercial jet engines are highly inefficient at low ground speeds, causing massive fuel waste and expensive Foreign Object Debris damage during taxi operations. This reliance on main engines generates severe air and noise pollution, directly threatening the health of ground crews and local communities. Ultimately, conventional taxiing creates a critical emissions bottleneck, preventing the aviation industry from achieving its zero-carbon and sustainable infrastructure goals.

Problem Understanding

- **Who experiences this problem?**
  Airlines (severe financial losses), airport operators (tarmac congestion), and ground crews and local communities (exposure to toxic exhaust and high-decibel noise).

- **What causes the problem?**
  Commercial jet engines are exclusively optimized for high-speed, high-altitude flight. Operating them at idle thrust for low speeds on the ground is inherently inefficient.

- **How does it affect users or stakeholders?**
  Airlines waste millions on aviation fuel and engine damage; ground workers face respiratory hazards; airports fail to meet national zero-carbon sustainability mandates.

Existing Gaps and Challenges

- **Conventional Jet and Tow-Tractor Gap:**
  Main engines waste massive fuel and emit CO2. Alternatives like tow-tractors severely clutter the apron and introduce manual connection bottlenecks.

- **On-Board Motor Gap:**
  Installing electric motors directly into landing gears adds permanent dead weight, drastically increasing airborne fuel consumption and reducing cargo payload.

- **Main unsolved challenge:**
  Achieving automated, zero-emission ground mobility without adding heavy machinery to the apron or permanent weight penalties to the aircraft itself.

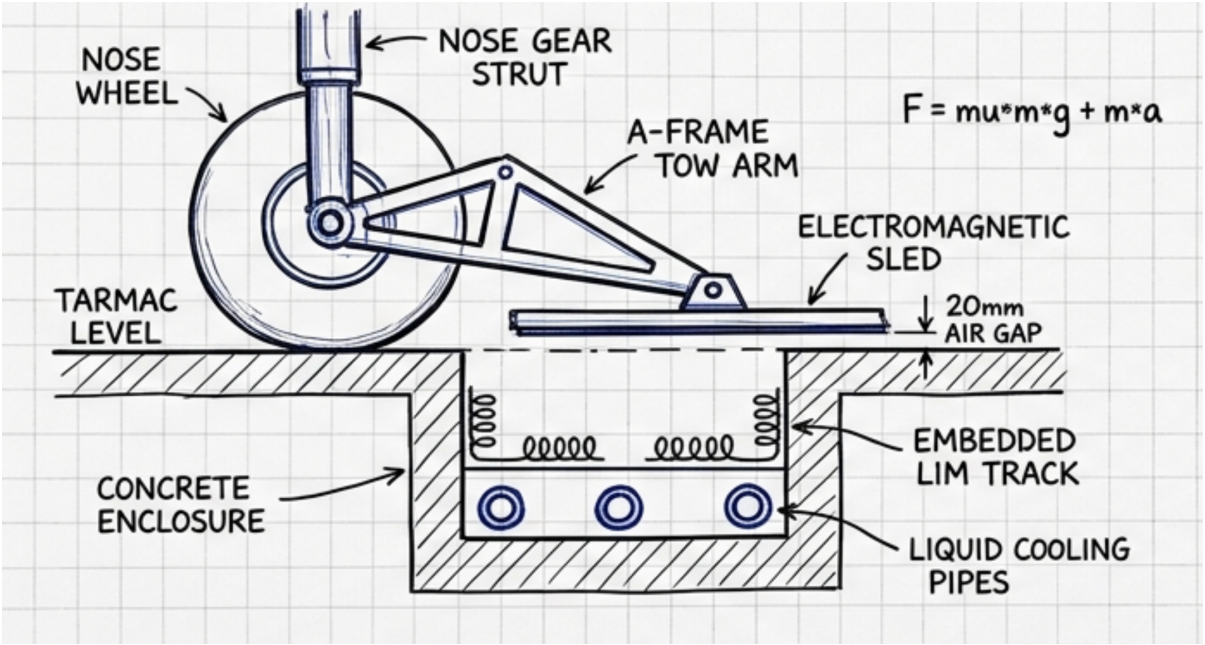
Our Proposed Solution

- **Core Idea: Project MagTaxi**
  A decentralized, zero-emission ground mobility network utilizing ultra-shallow Linear Induction Motors embedded directly into airport taxiways.

- **How the solution works at a high level:**
  The aircraft lowers a 50 kg passive receiver plate to a 20mm air gap. The track magnetically pulls the aircraft, while a Kinetic Energy Recovery System (KERS) brakes the plane and routes energy to gate battery banks.

- **Main value delivered to users:**
  Eliminates 90 percent of ground taxi jet fuel burn, protects against debris, guarantees zero jet blast around workers, and recovers megawatt-scale kinetic energy.

Technical Approach



The 4-inch stators generate thrust. Mu-Metal coatings shield the belly avionics. Approaching turns, software calculates maximum safe velocity and triggers KERS to brake automatically.

Tech Stack

Propulsion: 4-inch LIMs **Shielding:** Mu-Metal Coatings

Recovery: KERS Grid **Control:** Dynamic Cornering Logic

Innovation and Uniqueness

- **What is new or different?**
  Shifting the heavy propulsion mechanics out of the aircraft and into the smart infrastructure. We achieve this using an ultra-shallow 4-inch trench design that avoids sub-base drainage interference.

- **Why is this approach suited to the problem?**
  It perfectly balances civil engineering limits with aviation physics, utilizing Mu-Metal shielding to entirely eliminate radar and avionics interference.

- **What is the key differentiator?**
  Our dynamic cornering logic. The algorithm limits stress on the aircraft mass by automatically calculating cornering physics and utilizing free-castering to protect the nose gear.

Feasibility and Scalability

- **Feasibility:**
  The 4-inch track depth is highly feasible within standard top-layer Portland Cement Concrete, allowing phased, nighttime installation without disrupting sub-surface drainage.

- **Scalability:**
  Installation begins as pilot programs at smaller regional airstrips to validate the system before scaling up to high-density hubs and accommodating larger commercial aircraft.

- **Risks and Mitigations:**
  Risk: Engine thermal stabilization constraint. *Mitigation:* MagTaxi routes aircraft to dedicated warm-up holding zones 5 minutes before the runway to ensure safe takeoff thrust.

Cost for Product and Scalability

- **Scalable CapEx (Infrastructure and KERS Grid):**
  - *Small (1.5 to 2 km):* INR 60 to 100 Crore (Pilot phase)
  - ***Medium (3 to 5 km):*** **INR 125 to 250 Crore (Primary Target)**
  - *Large (10 to 15 km):* INR 450 to 750 Crore (Mega-hub phase)

- **Operational Cost Comparison (A320 aircraft):**
  Conventional 20-min taxi (320kg fuel at INR 171 per kg) = **INR 54,720**.
  MagTaxi electric traction (10 kWh) = **INR 100** (over 99 percent savings).

- **ROI and Payback (Medium Airport):**
  Eliminating millions in annual fuel and engine maintenance costs yields a projected ROI payback period of **3.5 to 5.0 years**.

Use Cases (compulsorily 3 use cases must be there)

- **Use case 1: The Commercial Pilot**
  Scenario: Upon landing, the pilot deploys the 50kg receiver and shuts down main engines. MagTaxi safely pulls the aircraft to the gate, eliminating 320kg of fuel burn and zeroing debris risk.

- **Use case 2: The Ground Support Worker**
  Scenario: Works around the terminal apron in complete silence, free from toxic jet exhaust and jet-blast hazards, utilizing recovered power from KERS for ground equipment.

- **Use case 3: The Airport Operations Manager**
  Scenario: Uses the MagTaxi central dashboard to schedule precise, automated ground routing, entirely eliminating apron congestion and taxiway holds.

Impacts and Benefits

- **Benefit to the target users:**
  Airlines drastically reduce their primary operating expense (aviation fuel) and increase engine lifespan by removing low-speed, high-stress ground operations.

- **Social, economic, or environmental impact:**
  Total elimination of localized ground-level CO2, NOx, and particulate matter emissions, paired with a massive reduction in occupational auditory and respiratory health hazards.

- **Expected measurable outcome:**
  Over 98 percent reduction in energy costs per ground movement cycle, alongside the generation of recoverable megawatt-scale electricity via the braking system.

Sustainable Development Goals (SDGs)

SDG No.	Target	Goal Name	Connection
SDG 3	Target 3.9	Good Health	Drastically reduces toxic emissions and noise for ground staff.
SDG 7	Target 7.2	Clean Energy	Kinetic recovery creates clean, localized gate microgrids.
SDG 9	Target 9.1	Infrastructure	Upgrades static concrete to intelligent transport corridors.
SDG 11	Target 11.6	Sustainable Cities	Improves localized air quality for communities near airports.
SDG 12	Target 12.2	Responsible Cons.	Minimizes wasted aviation turbine fuel during ground ops.
SDG 13	Target 13.2	Climate Action	Eliminates millions of tons of CO2 from high-density corridors.

Individual Member Contribution

Member	Contribution or responsibility
Varshith Vrindesh	Project Manager and Lead Systems Integrator
Jatin Dinesh	Infrastructure and Master Planning Lead
RVS Gurucharan	Operations and Reliability Engineer
Akshara Pulicherla	Ground Systems Engineer

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